Sri Lanka Institute of Information Technology



SE2030

Software Engineering

2026-Y2-S1-MLB-B3G1-08

Web-based Medicine/Drugs Ordering System

IT Number	**Name**
IT25101923	Lakshan M.K.C.
IT25102867	Karunathilaka W.V.R.D.T.Y.
IT25103946	Vinogikan M.
IT25101882	Widyawansa T.D.
IT25103725	Hathoon.M.F.S.
IT25100979	Kookulan E.

B.Sc. (Hons) in Information Technology

1. Agile sprint summaries

Sprint	Timeline	Focus Area and Key Deliverables	Core Artifacts and Modules Touched	Duration	Status
Spring 1	Week 4 - 7	• Prescription Upload & Clinical Verification Engine (Function 01) • Role-Based Access Control (RBAC) & Spring Security • FEFO Batch Allocation & Smart Inventory Management (Function 02) • Geofenced Delivery Management Baseline (Function 06)	• prescriptions & prescription_items schema/JPA • users RBAC claims & route guards • inventory_batches FEFO auto-allocation logic • deliveries entity & geofenced zone boundary baseline	54 hours	Completed
Sprint 2	Week 8 - 9	• Product Catalog UI & Usable Frontend Implementation • Role-Specific Admin Dashboards (Pharmacist, Delivery Coordinator, Operations) • Cold-Chain Packaging &	• React customer catalog & order creation forms • Privileged role dashboard portals & state guards • cold_chain_telemetry logging & courier checklist •is_chronic_subscription conditional retention flag • Tamper-proof append-only regulatory audit tables	62 hours	Completed

		Protocol Tagging (2°C – 8°C) • Chronic Care Refill & Long-Term Prescription Retention Engine • Financial Reconciliation Ledger & System Audit Trail (AOP) • Multi-Channel Notification Engine Foundations (Email, SMS, In-App)	• Multi-channel notification dispatch interfaces		
Sprint 3	Week 10 - 11	• Chronic Care Refill & Long-Term Prescription Retention Engine • Financial Reconciliation & System Audit Trail (AOP Interceptors) • Multi-Channel Notification Engine Foundations (Email, SMS, In-App) • Fleet Monitoring & Delivery Exception Handling	• is_chronic_subscription conditional retention flag • Tamper-proof append-only regulatory audit tables • Multi-channel notification dispatchers & gateway models • Live courier telemetry endpoints & transit delay alerts	58 hours	Partially Completed
Sprint 4	Week 12 - 14	• Secure Doorstep OTP Verification & 3-Attempt Lockout Handler • Operational Turnaround Time	• BCrypt OTP generation, courier keypad modal & photo POD • Operations & Financial reconciliation PDF/CSV export utilities • Payment webhooks, refund	50 hours	Planned

		(TAT) Analytics Dashboards • 3rd-Party Payment Gateway Integration (Stripe/Bank APIs) & COD Handover • Event-Driven Customer Alerts (Schedule, Expiry, Refill Reminders) • Administrative Security Auditing (Role updates, DB events, Traffic monitoring) • Notification Gateway Health Telemetry & Disaster Recovery Backups	handlers & cash audit workflow • Event-driven notification schedulers & message queues • Admin traffic monitor, DLQ retries & gateway heartbeats • Automated MySQL backup runbooks & point-in-time restore		

1. Completed Sprints

Sprint 1 (Weeks 4 – 7): Core Architecture, Prescriptions, Security & Inventory Baseline

1.1 Key Deliverables

- **Clinical Verification Engine Baseline (Function 01):** Secure multipart file ingestion pipeline capable of accepting digital prescription images (JPEG/PNG) and documents (PDF) with SHA-256 cryptographic hashing to prevent duplicate submissions.
- **Role-Based Access Control (RBAC) & Security Layer:** Integrated Spring Security architecture with JWT-based route guards and permission policies separating Customers, Pharmacists, Pharmacy Assistants, Delivery Coordinators, and Administrators.
- **FEFO Inventory Allocation Algorithm (Function 02):** Automated database query logic enforcing First-Expired-First-Out (FEFO) dispensing order, picking nearest-expiry drug batches before older stock.

- **Delivery & Geofence Boundary Schema (Function 06):** Initial database entities for deliveries and geofenced service zones, supporting address coordinate mapping and delivery status baselines.

1.2 Key Activities

- **Database Modeling & Schema Normalization:** Established core relational schema rules in MySQL (users, prescriptions, inventory_batches, deliveries), configuring JPA entity classes with Lombok to streamline persistence logic.
- **Cryptographic Hash Pipeline Implementation:** Developed backend hashing utilities in Spring Boot to compute file checksums during upload, preventing identical prescription re-uploads.
- **Security & Authentication Configuration:** Configured security filter chains, custom UserDetailsService, and role claims verification to restrict clinical queues exclusively to licensed personnel.
- **Batch Sorting Logic Development:** Authored custom Spring Data JPA repository queries to extract valid, non-quarantined batches ordered by expiry_date ASC during order allocation.
- **Unit & API Testing Execution:** Authored automated JUnit tests to validate the algorithmic accuracy of FEFO batch ordering, and established structured Postman API collections to test RBAC endpoint barriers, token validity, and multipart file upload payloads.

Sprint 2 (Weeks 8 – 9): Usable Frontend, Role Portals, Cold-Chain Safety & Expiry Controls

2.1 Key Deliverables

- **Interactive Frontend Product Catalog & Cart:** Responsive React.js customer storefront allowing users to browse over the counter (OTC) medicines, inspect drug details, filter categories, and stage cart items.
- **Privileged Role Dashboards:** Dedicated administrative interfaces tailored for Pharmacists (clinical prescription queue inspection) and Delivery Coordinators (dispatch logistics tracking).
- **Cold-Chain Protocol Tagging System (Function 04):** Implementation of temperature-sensitive medication flags across the product catalog and logistics dispatch manifests.

- **Automated Expiry Locking & Batch Quarantine:** Background verification mechanisms preventing expired or compromised medication batches from appearing in customer searches or cart checkouts.

2.2 Key Activities

- **React UI/UX Engineering:** Built reusable modular React components with Axios HTTP interceptors to handle authenticated communication with the Spring Boot REST API.
- **Inspection UI Development:** Engineered prescription preview components within the pharmacist portal, allowing side-by-side inspection of uploaded prescriptions and requested dosages.
- **Cold-Chain Workflow Integration:** Extended database entities with is_temperature_sensitive and requires_cold_chain flags, rendering clear visual warning badges across catalog views and courier packing lists.
- **Batch Quarantine Implementation:** Implemented server-side checks and status updates (ACTIVE $\rightarrow$ QUARANTINED / EXPIRED) to safeguard dispensing safety.
- **Integration & Regression Testing via Postman:** Executed comprehensive Postman test suites across all catalog filtering, batch locking, and role dashboard endpoints, confirming that non-privileged user roles receive HTTP 403 Forbidden responses.

2. Sprints in Progress

Sprint 3 (Weeks 10 – 11): Chronic Care Retention, Local Storage Pipeline & Notification Engine Foundations

3.1 Key Deliverables

- **Chronic Care Refill & Conditional Retention Engine (Function 05):** Data workflows supporting recurring 30-day refill intervals, including the is_chronic_subscription flag to retain prescriptions on disk while auto-deleting single-use files.
- **Local Storage & File System Persistence Architecture:** Dedicated local server storage subsystem for prescription PDFs and images, avoiding database bloat while maintaining strict file audit records in MySQL.
- **Regulatory System Audit Trail:** Tamper-proof, append-only logging tables recording all clinical approvals, inventory adjustments, and status transitions for compliance with NMRA standards.

- **Multi-Channel Notification Dispatcher Baseline:** Abstraction interfaces for SMS, Email, and In-App notification services designed with batching and rate-limiting controls to handle provider quotas.
- **Fleet Telemetry & Transit Exception Framework:** Courier tracking endpoints capable of capturing live coordinates and logging transit delays or customer unavailability.

3.2 Key Activities

- **File System & OS Permission Management:** Configuring Windows file system access rights, shell execution policies, and directory path normalizers to permit seamless read/write/delete operations for local prescription storage without OS-level permission blocks.
- **Conditional Physical File Purge Logic:** Implementing transactional deletion logic in Spring Boot to physically purge single-use prescription files from the server disk upon pharmacist approval while preserving database audit metadata.
- **Notification Rate-Limiting & Gateway Adaptation:** Designing asynchronous queues and throttling mechanisms to manage external API tier limits and mitigate costs when broadcasting transactional notifications to multiple users.
- **Aspect-Oriented Audit Interceptor Setup:** Configuring Spring AOP @Around / @AfterReturning interceptors to automatically capture user IDs, timestamps, and state changes on critical operations.
- **Cloud Deployment Pre-Requisites Evaluation:** Investigating cloud containerization configurations (Docker) to resolve environment variable and disk persistence discrepancies experienced during initial cloud deployment attempts.

3. Upcoming Sprints

Sprint 4 (Weeks 12 – 14): Handover Verification, Stripe Gateway Integration, Analytics & Disaster Recovery

4.1 Planned Deliverables

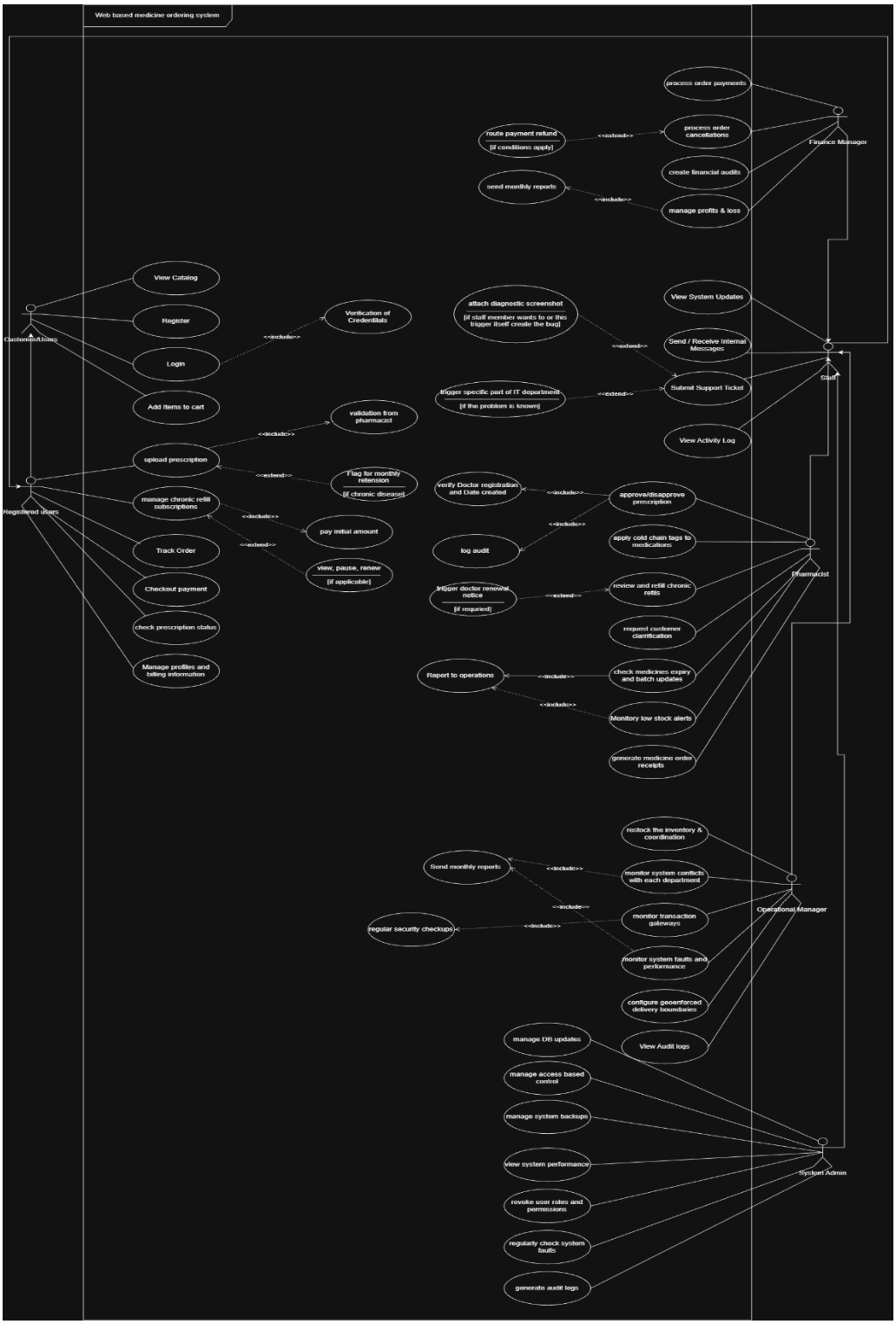
- **Secure Doorstep OTP Handover & 3-Attempt Lockout (Function 06):** Dynamic 6-digit OTP generation upon courier arrival, encrypted verification, 3-attempt brute-force protection, and proof-of-delivery (POD) photo capture.
- **Stripe Payment Gateway Integration & Cash-On-Delivery (COD) Balancing:** Complete card checkout processing using Stripe API (Stripe Elements / Checkout Sessions), asynchronous payment webhook handling, and courier COD cash reconciliation ledger.

- **Operations Turnaround Time (TAT) Analytics Dashboards:** Operational dashboards reporting average prescription review times, fulfillment durations, and delivery fleet turnaround times.
- **Notification Gateway Health Monitoring & Automated Backups:** Heartbeat monitoring for third-party gateways (SMS/Email), Dead Letter Queue (DLQ) retry mechanisms, and automated daily MySQL database snapshot routines.

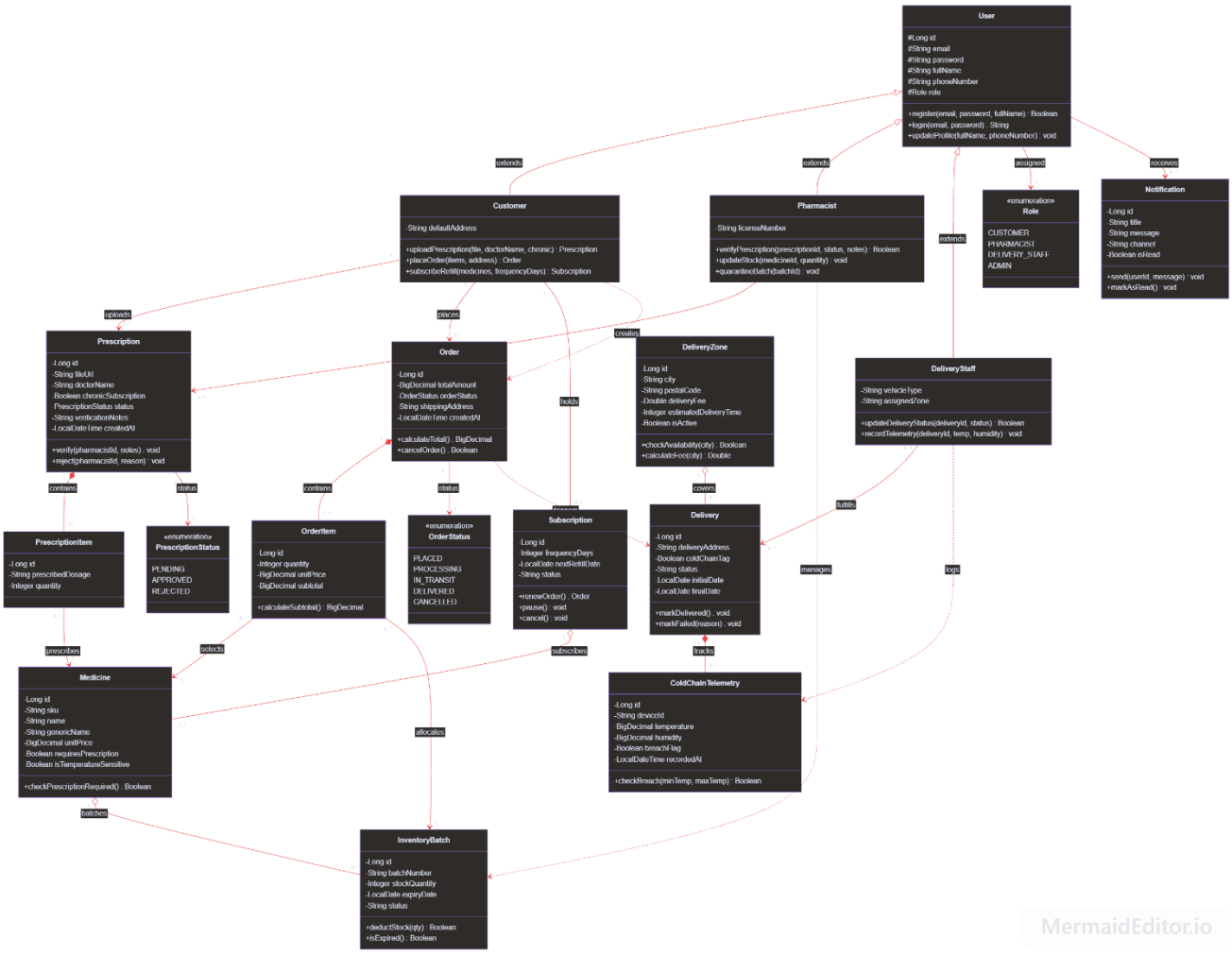
4.2 Planned Activities

- **Cryptographic OTP Verification Engineering:** Implementing BCrypt OTP hashing upon dispatch, building a courier keypad entry modal, and enforcing account lockouts after 3 invalid attempts.
- **Stripe Webhook & Transaction Flow Implementation:** Connecting backend endpoints to Stripe webhooks (payment_intent.succeeded, charge.refunded) to advance order states automatically upon successful payment.
- **Operational Reporting & SQL Aggregations:** Authoring optimized SQL aggregation queries and charts to display business KPIs, fulfillment bottlenecks, and stock turnover metrics.
- **Disaster Recovery Simulation:** Testing point-in-time database restoration routines and verifying system resilience under simulated network disconnects.

2. Final use case diagram

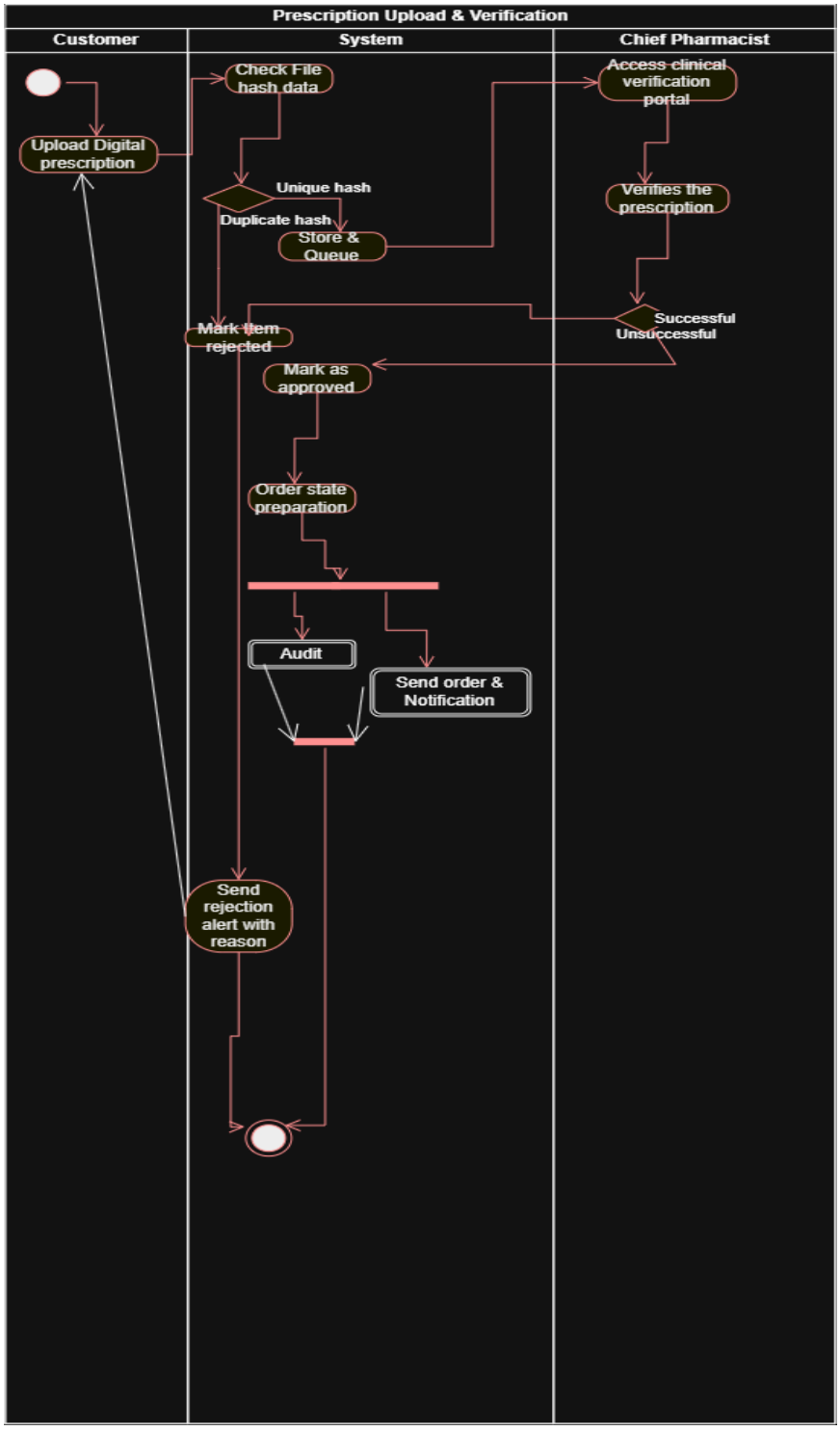


3. Class diagrams

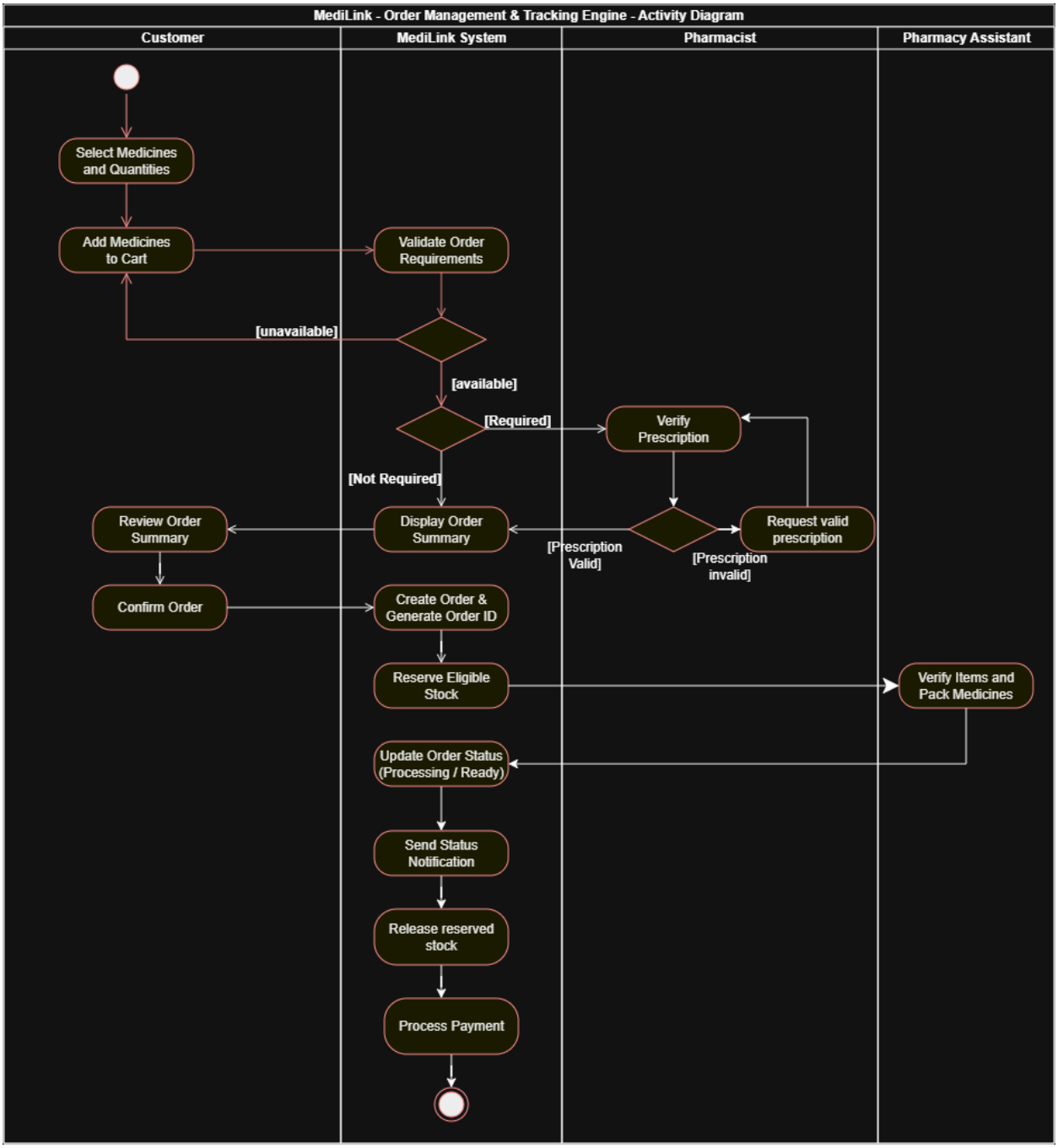


4. Activity diagrams

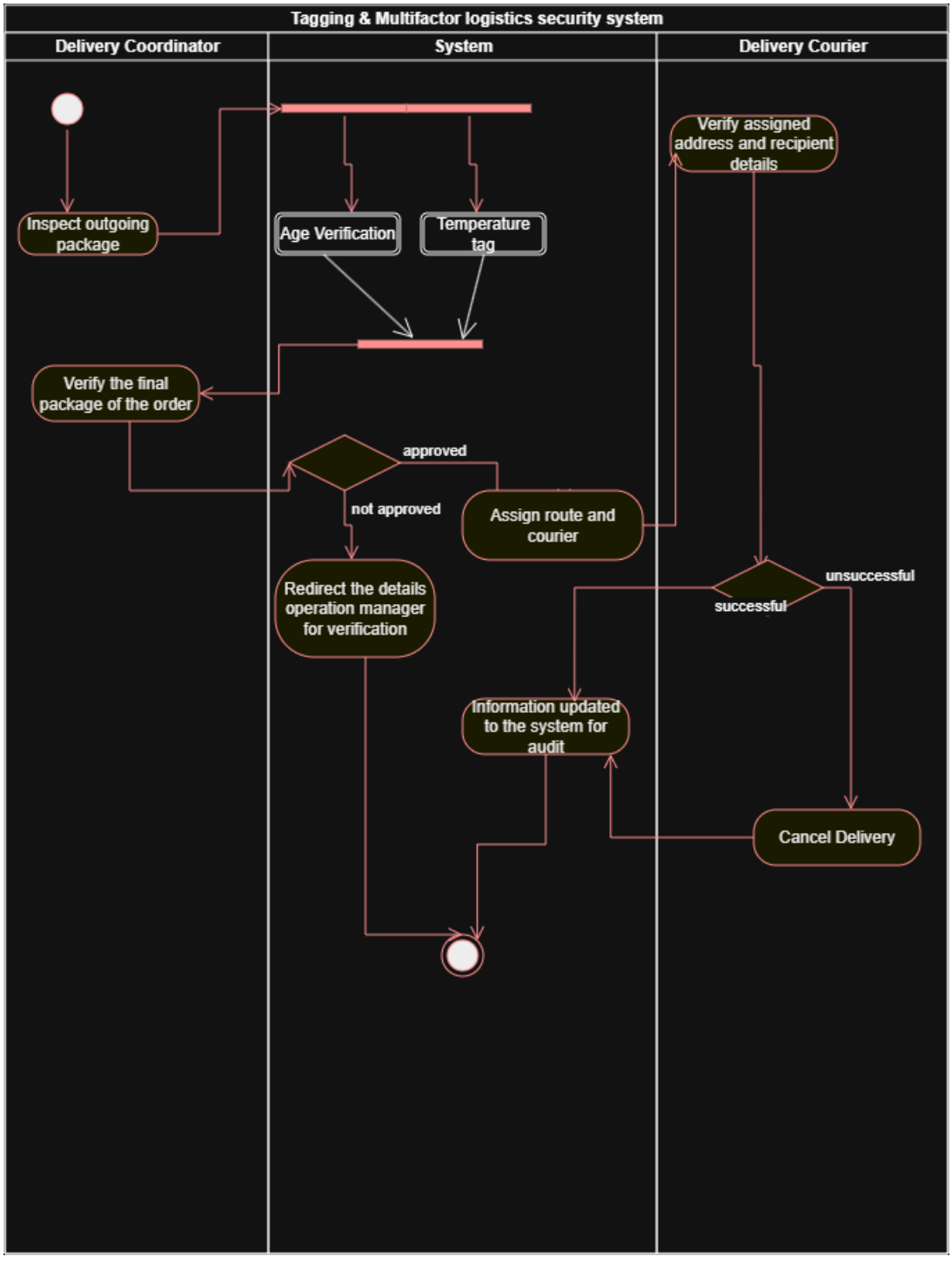
4.1 Prescription upload & clinical verification



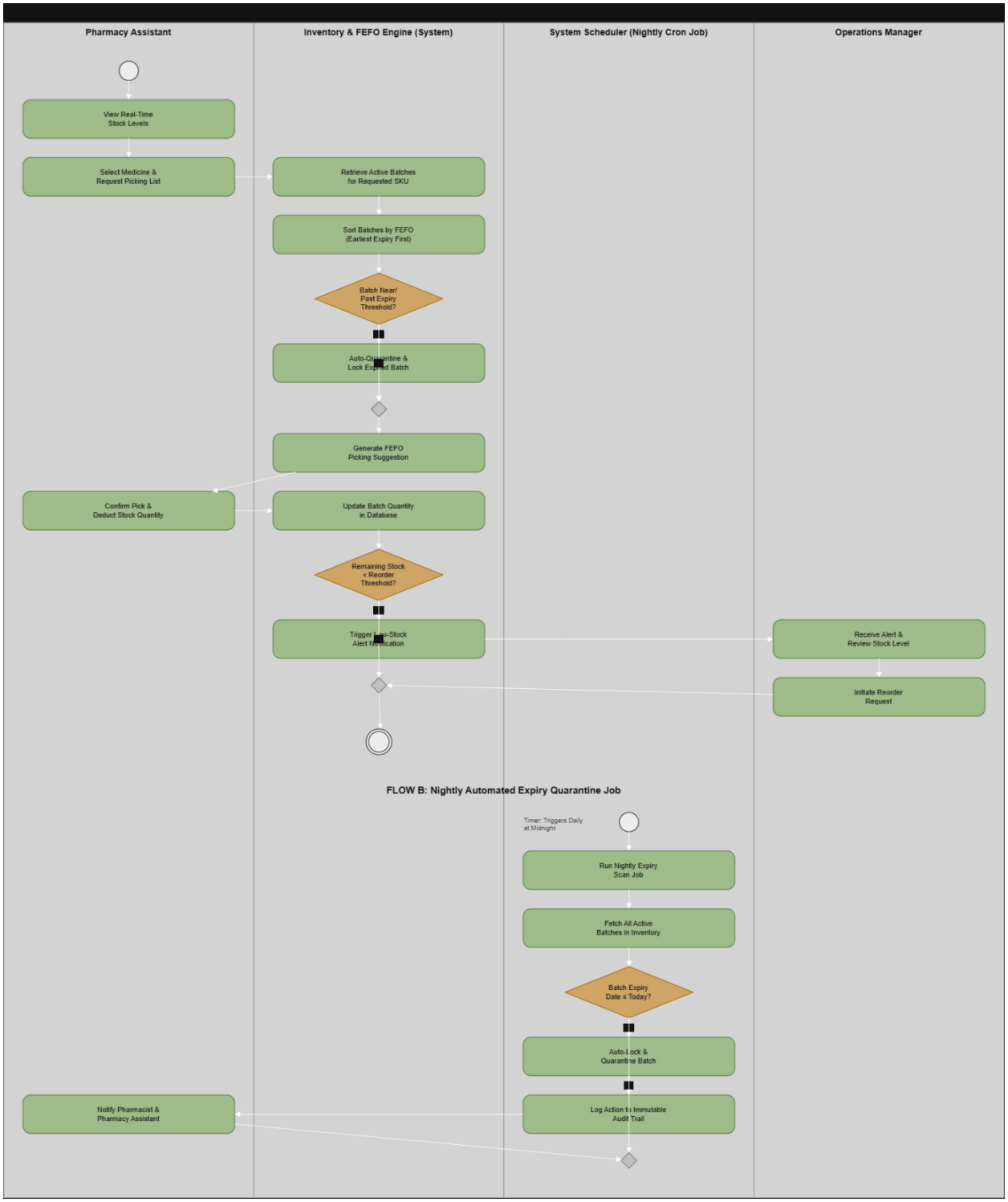
4.2 Inventory & FEFO expiry management



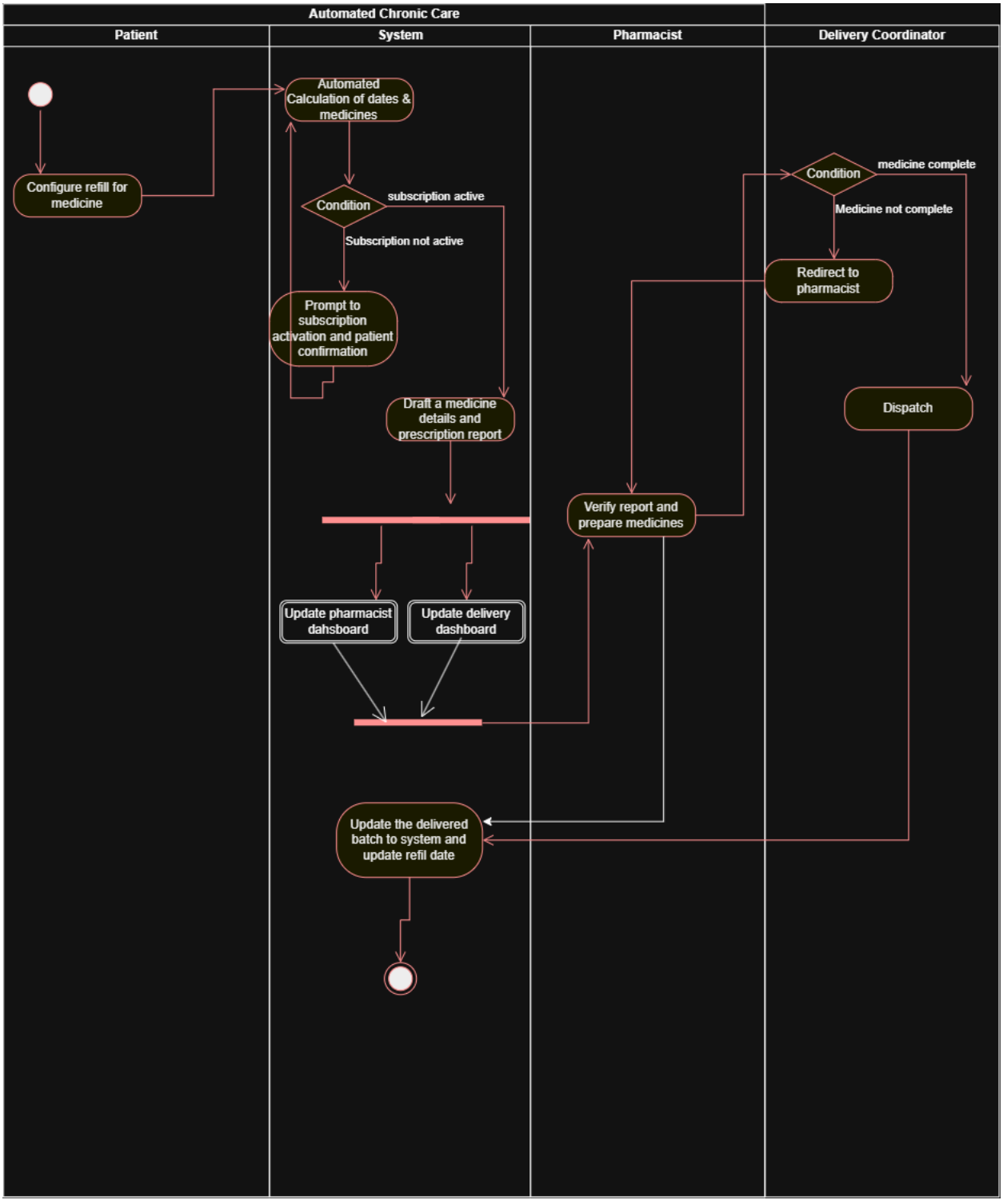
4.3 Cold chain and multi factor logistics system



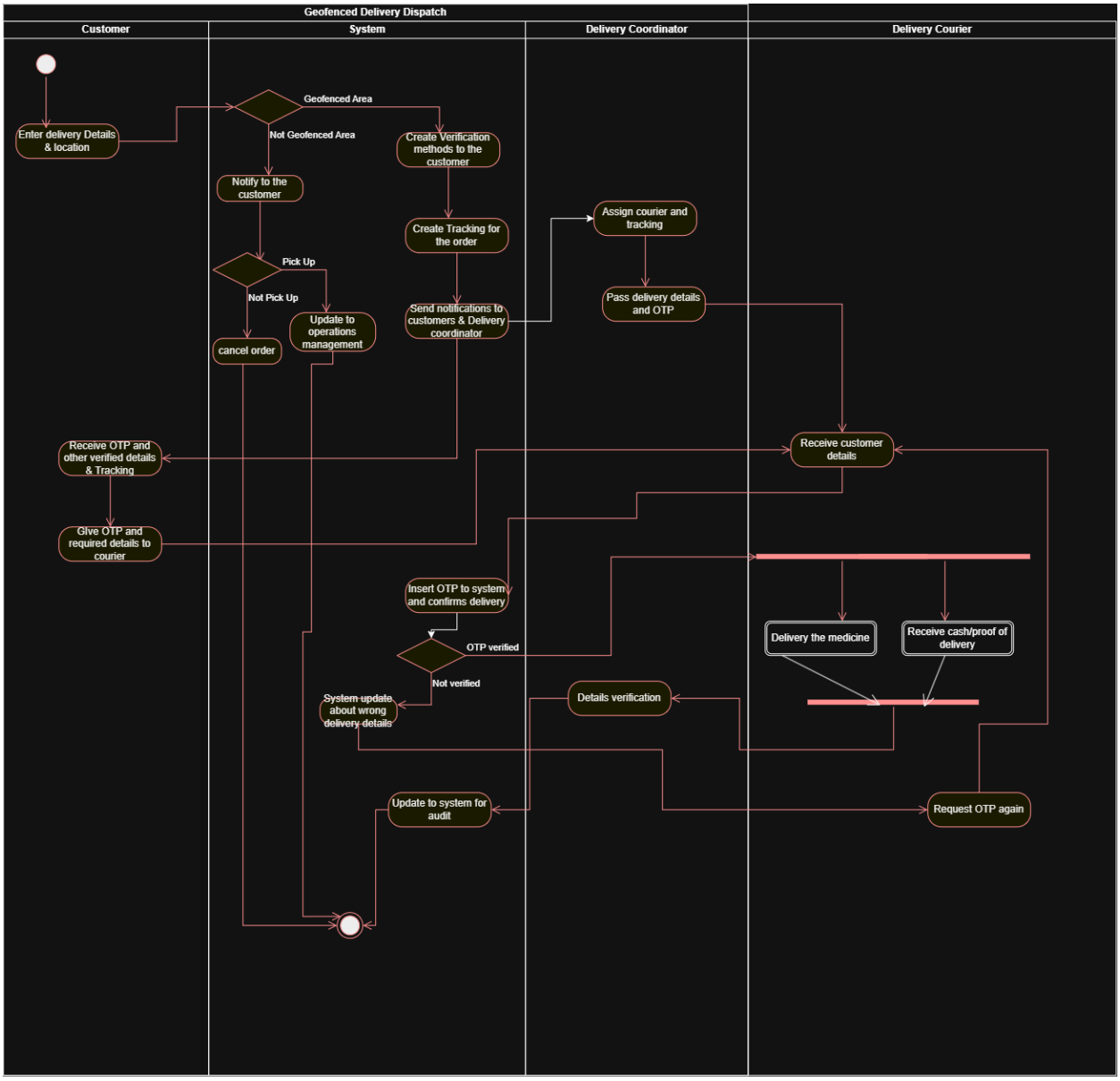
4.4 Real Time order processing & state workflow tracking



4.5 Automated chronic care refill & subscription management



4.6 Geofenced delivery dispatch system & multichannel notification



5. Ethical Considerations

5.1 Data Privacy and Confidentiality

Description of Concern

Digital pharmacy systems process sensitive personal and medical data, including patient identities, diagnostic information, prescribed medication regimens, home addresses, and clinician details. Unauthorized exposure, persistent leakage, or mishandling of digital prescriptions constitutes a severe violation of medical confidentiality. Traditional database storage of raw patient documents introduces vulnerabilities to data harvesting, unauthorized internal browsing, and regulatory non-compliance with data protection standards.

How We Address It: -

- **Automated Ephemeral File Purging:** Single-use prescription image and PDF files stored on the local filesystem are programmatically deleted immediately following clinical verification, clearing the active file path from the database to eliminate unnecessary digital retention.
- **Consent-Driven Chronic Retention:** Physical document retention is permitted only when a patient explicitly selects and authorizes recurring monthly refills (is_chronic_subscription), ensuring storage strictly aligns with informed user consent.
- **Strict Role-Based Access Control (RBAC):** Backend endpoints and static resources are secured using Spring Security, restricting direct access to prescription documents exclusively to verified clinical staff (pharmacists and system administrators) while blocking unauthorized users and public access.
- **Cryptographic Data Integrity:** Document deduplication relies on SHA-256 cryptographic hashing to detect duplicate uploads without exposing raw medical file contents across shared channels.

5.2 Accountability and Accuracy

Description of Concern

Dispensing pharmaceuticals through an online system carries substantial safety risks, including misinterpretation of blurry or unverified prescriptions, unauthorized dispensing of restricted drugs, dispensing expired batches, and fraudulent doorstep deliveries. In an automated environment, the lack of an attributable audit trail obscures liability between the prescribing doctor, the reviewing pharmacist, and the dispatch courier if adverse medication events or fulfillment errors occur.

How We Address It: -

- **Doctor Credential Verification:** The pharmacist interface requires verification of the prescribing doctor's identity and Sri Lanka Medical Council (SLMC) registration credentials prior to approving any controlled prescription.
- **Tamper-Proof Audit Logging:** Every verification decision records an immutable audit record in the database capturing the reviewing pharmacist's unique identifier (verified_by), the precise verification timestamp (verified_at), the clinical decision, and mandatory clinical rejection remarks.
- **FEFO Inventory Safeguards:** Automated First Expiry, First Out (FEFO) batch allocation algorithms and automated quarantine locks prevent the dispatch of expired or near-expiry medicines.
- **Doorstep OTP Handover Verification:** Deliveries are secured with a mandatory, one-time verification code (OTP) presented by the registered recipient at the doorstep, ensuring accountable chain-of-custody transfer and preventing misdelivery of critical pharmaceuticals.